

Formaldehyde

CAS 50-00-0 · DTXSID7020637 · CH₂O

Toxicological Literature Review

A chemical-agnostic, multi-database literature synthesis

Prepared for: SciToVation

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Databases surveyed: 37 · Records identified: 1,828,641

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Executive Summary

Formaldehyde (CAS 50-00-0; IUPAC methanal; formula CH_2O ; molecular weight 30.026 g/mol) is a colourless, pungent, highly reactive one-carbon aldehyde that is produced endogenously as an intermediate of single-carbon metabolism in every living cell and also manufactured on an enormous scale for industrial use. Because it is simultaneously an obligate endogenous metabolite and one of the most widely manufactured chemicals in the world, formaldehyde sits at the centre of decades of contested risk-assessment debate about the shape of the human dose-response curve for cancer and non-cancer respiratory endpoints. The present literature review consolidates evidence retrieved from a parallel sweep of 37 scientific and regulatory databases conducted on 21 April 2026 and synthesises the findings against the most recent authoritative assessments, principally the United States Environmental Protection Agency (EPA) Integrated Risk Information System (IRIS) Toxicological Review of Formaldehyde — Inhalation (final August 2024), the National Toxicology Program (NTP) 15th Report on Carcinogens (2021), the Agency for Toxic Substances and Disease Registry (ATSDR) Toxicological Profile tp111 (1999, 468 pp), the World Health Organization (WHO) Environmental Health Criteria monograph 89 (1989), and the National Institute for Occupational Safety and Health (NIOSH) Pocket Guide to Chemical Hazards (entry npgd0293, reviewed 30 October 2019).

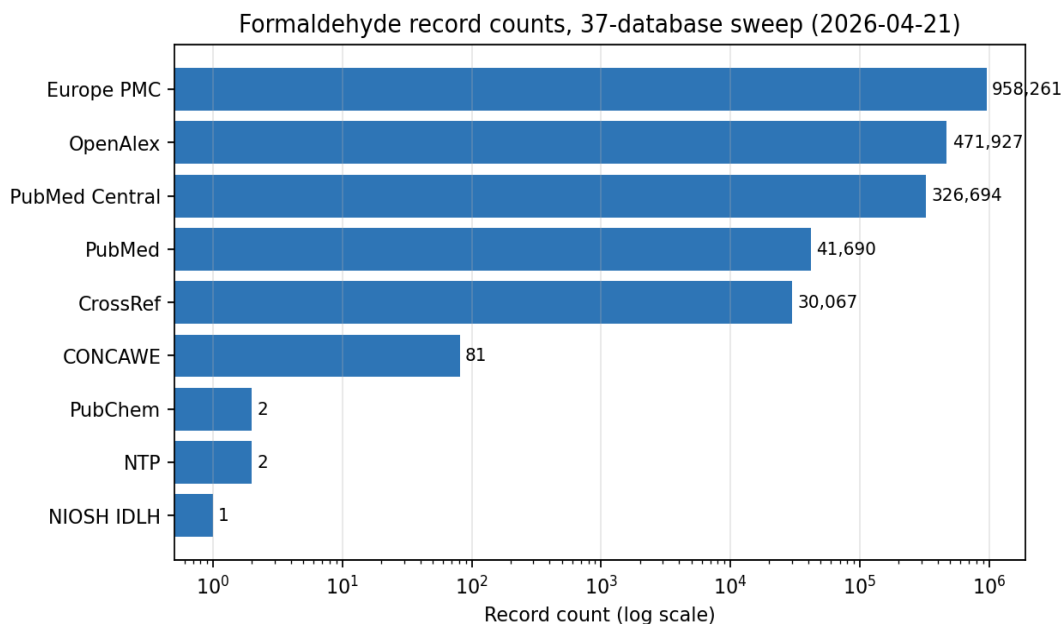


Figure 1. Formaldehyde record counts across the nine databases with explicit record returns in the 21 April 2026 parallel sweep. Counts shown on log scale.

Across the 37 databases queried, 1,828,641 primary-source records referencing formaldehyde were identified, with the largest single contributors being Europe PMC (958,261 records), OpenAlex (471,927), PubMed Central (326,694), PubMed (41,690), and CrossRef (30,067). An additional 84 records were retrieved directly from agency HTML resources including NTP profiles, the CONCAWE publications portal,

and the NIOSH IDLH documentation. Seventeen peer-reviewed studies of particular regulatory and mechanistic significance were catalogued in full bibliographic detail and are discussed throughout this review. The full database-by-database coverage is provided in Appendix A.

Quantitatively, the current health-protective benchmarks for formaldehyde by inhalation are dominated by the EPA IRIS 2024 final values. EPA derived a chronic inhalation reference concentration (RfC) of 0.007 mg/m³ (approximately 5.6 parts per billion), identified as the median of three organ-specific RfCs capturing (i) rhinoconjunctivitis in children (0.008 mg/m³, Annesi-Maesano 2012), (ii) decreased peak expiratory flow rate indicating pulmonary-function loss (0.007 mg/m³, Krzyżanowski 1990), and (iii) worsened asthma control (0.006 mg/m³). The noncancer oral reference dose (RfD) is 0.2 mg/kg-day, unchanged from the 1990 IRIS assessment and based on reduced weight gain and histopathology in rats at a NOAEL of 15 mg/kg-day with a composite uncertainty factor of 100. EPA concludes that formaldehyde is “carcinogenic to humans by the inhalation route of exposure” under the 2005 Guidelines for Carcinogen Risk Assessment, on the basis of three independent weight-of-evidence judgments for nasopharyngeal cancer, sinonasal cancer, and myeloid leukemia. A unit inhalation cancer risk (IUR) of 1.1×10^{-5} per µg/m³ is derived from the Beane Freeman (2013) update of the National Cancer Institute industrial cohort, with age-dependent adjustment factors (ADAFs) applied in recognition of the operant mutagenic mode of action for nasal tumours. Occupationally, NIOSH treats formaldehyde as an Appendix A potential human carcinogen and recommends a lowest-feasible TWA of 0.016 ppm with a 15-minute ceiling of 0.1 ppm; the OSHA PEL of 0.75 ppm TWA (29 CFR 1910.1048) is two orders of magnitude above the NIOSH REL and its supporting science predates the current cancer determination. The NIOSH IDLH is 20 ppm.

The NTP 15th Report on Carcinogens (2021) continues to list formaldehyde as “known to be a human carcinogen,” a designation upgraded in the 12th RoC (2011) from the “reasonably anticipated to be a human carcinogen” status it had held since its original listing in the Second Annual Report on Carcinogens (1981). The NTP listing rests on a causal relationship with three tumour sites: nasopharyngeal cancer (supported by the Vaughan 2000 case-control study reporting an odds ratio of 13.3, 95% CI 2.5–70, among subjects with definite occupational exposure), sinonasal cancer (supported by the Luce 2002 pooled analysis of twelve case-control studies), and myeloid leukemia (supported by the Beane Freeman 2009 update of the NCI industrial cohort of over 25,000 workers, the Hauptmann 2009 NCI embalmers case-control study, and the Zhang 2009 meta-analysis).

Mechanistically, the past five years have clarified the enzymatic systems that detoxify endogenous formaldehyde and the cellular catastrophes that ensue when they fail. Mulderrig et al. (2024, *Nucleic Acids Research*) documented how distinct DNA repair pathways — homologous recombination, nucleotide excision repair, and the Fanconi anaemia pathway — act in concert to prevent the lethal and mutagenic DNA-protein crosslinks (DPCs) that formaldehyde produces. Niu et al. (2025, *Leukemia*) showed that the ADH5/ALDH2 dehydrogenase pair, together with DNA polymerase θ (Polθ), protects both normal and malignant hematopoietic cells from formaldehyde challenge, with direct therapeutic implications for Fanconi anaemia and AML patients. Fujihara et al. (2024, *Nature Cell Biology*)

demonstrated that transcription-coupled nucleotide excision repair (TC-NER) resolves endogenous aldehyde-induced DPCs, explaining long-standing clinical observations of neurodegeneration and premature aging in AMeD (aplastic anaemia, mental retardation, and dwarfism) syndrome patients who carry homozygous ALDH2 loss-of-function alleles.

Epidemiologically, the Protano et al. (2022, *Cancers*) systematic review of occupational exposure confirmed the three NTP cancer sites, while the Hauptmann (2009) embalmer cohort reported a monotonic dose-response for myeloid leukemia. The Wang (2015, *Scientific Reports*) pilot study reported reduced semen quality in occupationally exposed workers, and the Yang (2021, *Neurotoxicity Research*) meta-analysis and bioinformatics reanalysis linked chronic formaldehyde exposure to Alzheimer's-type neurodegeneration through tau hyperphosphorylation and endogenous formaldehyde dysregulation. More recent observational studies including Chu (2025, *IJOMEH*) on low-dose occupational health effects and the Korean allergic rhinitis mouse model (2026) reinforce the IRIS 2024 respiratory endpoints.

Taken together, the post-2020 mechanistic and epidemiological evidence reinforces rather than qualifies the regulatory trajectory initiated by the EPA 2024 RfC reduction. The remainder of this report surveys, chapter by chapter, the physicochemical identity of the compound, its production and uses, environmental fate and human exposure patterns, toxicokinetics and metabolism, acute and repeated-dose toxicity, carcinogenicity, genotoxicity, reproductive and developmental effects, and regulatory limits; the whole is anchored on the live 2026 database sweep summarised in Appendix A and the seventeen primary-source studies catalogued in the References.

1. Chemical Identity and Physicochemical Properties

1.1 Nomenclature, registry identifiers, and structural formula

Formaldehyde is the simplest of the aldehydes and the parent compound of the homologous series $\text{H}(\text{CH}_2)_n\text{CHO}$ with $n=0$. The International Union of Pure and Applied Chemistry (IUPAC) preferred name is methanal; the name "formaldehyde" is retained in common usage and throughout regulatory and industrial practice. Additional synonyms listed in the NIOSH Pocket Guide include methyl aldehyde and methylene oxide; when dissolved in water at 37 percent weight/volume with 10–15 percent methanol as a stabiliser it is traded as formalin, the form in which most aqueous formaldehyde is handled in laboratories and histology suites. Chemical Abstracts Service registry number 50-00-0 is the universal identifier in chemical and toxicological databases; the EPA DSSTox DTXSID7020637 is the EPA CompTox Chemicals Dashboard identifier and is used internally by the Integrated Risk Information System (IRIS). The PubChem CID 712 indexes the compound in the NIH small-molecule library. The simplest molecular representation is CH_2O , expanded as $\text{H}-\text{C}(=\text{O})-\text{H}$; the InChI key is WSFSSNUMVMOOMR-UHFFFAOYSA-N.

1.2 Physicochemical properties

Property	Value
Formula	CH_2O (methanal)
CAS number	50-00-0
DTXSID	DTXSID7020637
PubChem CID	712
Molecular weight	30 g/mol
Physical description	Nearly colorless gas with a pungent, suffocating odor. Often used in aqueous solution (Formalin).
Boiling point	-6 °F (-21 °C)
Freezing point	-134 °F (-92 °C)
Vapour pressure (20 °C)	>1 atm
Ionization potential	10.88 eV
Lower explosive limit (LEL)	7%
Upper explosive limit (UEL)	73%
Relative gas density (air=1)	1.04
Solubility in water	Miscible
Flammability	Flammable gas (pure); aqueous formalin is combustible

Property	Value
Odour threshold	≈ 0.83 ppm in air (subjective)
Incompatibilities	Strong oxidizers, alkalis & acids; phenols; urea. (Pure FA polymerizes; reacts with HCl to form bis-chloromethyl ether.)

The physical properties relevant to occupational hygiene reflect formaldehyde's status as a gas at ordinary temperatures that is nonetheless typically handled in aqueous solution. The extremely low boiling point means that open-topped formalin containers release formaldehyde vapour continuously; the very wide flammability range (7–73 percent by volume) is a fire-safety consideration in manufacturing environments, although dilute aqueous solutions below 50 percent are not classified as flammable liquids under 29 CFR 1910.106. Pure formaldehyde has a strong tendency to polymerise to paraformaldehyde ($\text{HO}(\text{CH}_2\text{O})_n\text{H}$) on standing, a process accelerated by trace water and retarded by methanol addition. Hazardous secondary products include bis-chloromethyl ether (BCME), formed when formaldehyde reacts with hydrochloric acid or concentrated chloride brines; BCME is itself a potent human lung carcinogen and its inadvertent formation in industrial settings has historically driven several of the most stringent engineering controls for formaldehyde handling.

1.3 Endogenous formation

Uniquely among widely regulated industrial carcinogens, formaldehyde is produced inside every living cell at substantial concentrations as an obligate metabolic intermediate. Endogenous sources include one-carbon demethylation reactions on DNA (catalysed by the Fe(II)/ α -ketoglutarate-dependent dioxygenase family), demethylation of histones, oxidation of dietary and endogenous methanol by alcohol dehydrogenase 5 (ADH5, also called S-hydroxymethyl-glutathione dehydrogenase or formaldehyde dehydrogenase), cytochrome P450 N-demethylation of xenobiotics, and the spontaneous decay of 5-formylcytosine in genomic DNA (Mulderigg 2024). Steady-state tissue formaldehyde concentrations have been measured in the micromolar range in whole blood (approximately 70–100 μM), in brain cerebrospinal fluid, and in saliva (Swenberg 2013). The implication is that any external exposure must be evaluated against a large endogenous baseline, and that standard mass-action toxicokinetic models break down at the low-dose end where exogenous formaldehyde contributes only a small increment to the pool of formaldehyde-derived DNA adducts (N^2 -hydroxymethyl-dG) that are detectable in unexposed individuals (Swenberg 2013). This endogenous floor is central to the interpretation of the linear low-dose cancer model adopted by EPA in the 2024 IRIS assessment.

2. Production, Commerce, and Uses

2.1 Industrial synthesis

Commercial formaldehyde is produced almost exclusively by the catalytic oxidation of methanol, either by the "silver process" (partial oxidation over a silver crystal catalyst at approximately 600–650 °C with excess methanol) or the "Formox process" (complete oxidation over an iron-molybdenum-oxide catalyst at 300–400 °C with excess air). The silver route yields a product stream of roughly 37 percent formaldehyde with residual methanol, which is suitable for stabilised formalin; the Formox route produces a substantially methanol-free 55 percent aqueous solution that is preferred for downstream resin manufacture. Both routes are highly exothermic and are typically operated with integrated heat recovery. The World Health Organization Environmental Health Criteria monograph 89 (1989) described world production capacity in the late 1980s at approximately 18 million tonnes per annum on a 37 percent weight/volume basis, corresponding to approximately 6.7 million tonnes on a 100 percent formaldehyde basis; current global production is estimated by trade sources at 30–40 million tonnes per annum on a 37 percent basis, reflecting continued growth in wood-panel and engineered-plastic markets. Over 50 percent of global production is consumed captively at the production site for immediate downstream polymerisation into urea-formaldehyde, phenol-formaldehyde, and melamine-formaldehyde resins.

2.2 End uses

The single largest end-use of formaldehyde globally is the manufacture of amino resins, predominantly urea-formaldehyde (UF) and melamine-urea-formaldehyde (MUF) binders used in the production of particleboard, medium-density fibreboard (MDF), oriented strand board (OSB), plywood, and other engineered wood panels. These composite-wood products account for an estimated 55–65 percent of global formaldehyde consumption depending on the reference year and source. Phenol-formaldehyde resins are the second-largest use and are preferred for exterior-grade and structural plywood, laminated veneer lumber, and industrial adhesives because of their superior hydrolytic stability. Other major derivatives include polyoxymethylene (POM) engineering plastics used in automotive, electrical, and consumer goods; 1,4-butanediol used as a feedstock for tetrahydrofuran and polybutylene terephthalate; methylenedianiline (MDA) used as the hardener in epoxy systems and the precursor for MDI polyurethanes; hexamethylenetetramine (HMT or urotropine); pentaerythritol used in alkyd resins and explosives; and trimethylolpropane and trimethyloethane used as crosslinkers.

Beyond the polymer market, formaldehyde is used directly as a biocide in paints, personal-care products, and metalworking fluids (typically as a release-agent such as quaternium-15 or DMDM hydantoin that liberates free formaldehyde in use); as a tissue fixative in pathology and anatomy (generally as 10 percent neutral buffered formalin, delivering approximately 3.7 percent formaldehyde); as an embalming agent; and as a sterilant for medical instruments where heat or ethylene oxide are unsuitable. Historical uses in textile finishing (permanent-press treatments) and in cosmetic hair-smoothing systems ("Brazilian

blowout") have been progressively restricted on health-protective grounds in the European Union, California, and several other jurisdictions.

2.3 Trade and regulatory classifications affecting commerce

Formaldehyde is listed on the EPA Toxic Release Inventory (Section 313 of EPCRA) with a reporting threshold that triggers annual disclosure for manufacturing, processing, and other-use quantities exceeding 25,000, 25,000, and 10,000 pounds respectively. It is a HAP (hazardous air pollutant) under CAA Section 112. It is restricted under California Proposition 65 for both cancer and developmental toxicity. Under EU REACH it is included in the Candidate List of Substances of Very High Concern and in the Annex VI harmonised classification as Carc. 1B, Muta. 2, Acute Tox. 3, Skin Corr. 1B, and Skin Sens. 1. Several jurisdictions including Canada (Priority Substances List Assessment), Australia (Industrial Chemicals Environmental Management Standard), Japan (PRTR), and South Korea (K-REACH) maintain active national-level controls. The ban on composite-wood products exceeding emission limits in California (CARB ATCM for Composite Wood Products) and the federal counterpart (EPA TSCA Title VI rule, 40 CFR 770) have driven a generational shift toward no-added-formaldehyde (NAF) and ultra-low-emitting formaldehyde (ULEF) binder systems.

3. Environmental Fate and Human Exposure

3.1 Sources to air

Ambient air concentrations of formaldehyde arise from both primary anthropogenic emissions and secondary photochemical formation from the oxidation of methane, isoprene, terpenes, alkenes, and oxygenated hydrocarbons of both biogenic and anthropogenic origin. The WHO Environmental Health Criteria monograph 89 (1989) and subsequent updates identify motor vehicle exhaust, residential wood combustion, open biomass burning, photochemical smog, and industrial point sources (formaldehyde manufacturing, resin production, and embalming services) as the principal primary contributors in urban and peri-urban airsheds. Typical urban outdoor concentrations range from 1 to 20 $\mu\text{g}/\text{m}^3$ (approximately 0.8–16 ppb) with diurnal and seasonal variation tracking photochemical activity; rural background levels are generally an order of magnitude lower.

3.2 Indoor air

Indoor concentrations typically exceed outdoor levels by factors of 2–10 and are dominated by emission from composite wood products with urea-formaldehyde binders (new particleboard furniture, MDF cabinetry, engineered-wood flooring), by latex and acrylic paints containing formaldehyde-releasing biocides, by consumer products containing formaldehyde donors, by environmental tobacco smoke, and by combustion appliances including gas ranges and unvented kerosene heaters. Historical indoor concentrations in manufactured housing with urea-formaldehyde foam insulation (UFFI) were documented in the 1980s at levels exceeding 1,000 $\mu\text{g}/\text{m}^3$ in some residences, prompting the CPSC ban on UFFI and the CARB/TSCA Title VI product-emission rules. Present-day median indoor levels in new homes constructed under the California ATCM regime are in the 10–50 $\mu\text{g}/\text{m}^3$ range, still above most outdoor concentrations and frequently at or above the EPA IRIS 2024 RfC of 7 $\mu\text{g}/\text{m}^3$. The Morales (2025) Puerto Rico monitoring study documented elevated indoor formaldehyde in hurricane-damaged housing stock with continuous emission from replaced composite wood materials.

3.3 Water, soil, and food

Formaldehyde is present in surface waters at low nanogram-per-litre concentrations arising from atmospheric deposition and direct effluent from resin manufacturing facilities; it does not bioaccumulate and is rapidly metabolised by aquatic microorganisms and aerobic soil bacteria via the methylglyoxal pathway. Drinking water concentrations are of minimal public-health significance except where ozone treatment generates formaldehyde as a disinfection by-product, in which case typical concentrations remain below 100 $\mu\text{g}/\text{L}$. Endogenous formaldehyde is present in many foods including fresh fruits (especially apples and pears), aged dairy (some cheeses at 10–50 mg/kg), shellfish (a metabolite of trimethylamine-N-oxide), coffee, and fermented beverages. The Mohamed (2025) review summarises food-processing reactions (Maillard browning, lipid oxidation, smoke exposure) that generate

formaldehyde in finished foods, although the endogenous contribution from amino-acid demethylation greatly outweighs typical dietary intake.

3.4 Occupational exposure

The occupational settings with the highest documented formaldehyde exposures are anatomical pathology laboratories, embalming facilities in funeral homes, histology laboratories, foundries using phenol-formaldehyde sand binders, particleboard and MDF manufacturing, urea-formaldehyde resin manufacturing, and some textile and paper-product finishing operations. Historical industrial hygiene surveys conducted by NIOSH and IARC working-group teams documented eight-hour time-weighted average concentrations in the 0.1–5 ppm range in these industries between 1960 and 1990, with short-duration task exposures (fixing a cadaver, injecting embalming fluid, opening a press) frequently reaching 5–25 ppm. Progressive engineering controls (downdraft tables, enclosed embalming systems, press enclosures) and the reduction of the OSHA PEL to 0.75 ppm in 1992 have reduced typical exposures in developed economies by approximately one order of magnitude, but documented excursions above the NIOSH REL of 0.016 ppm remain routine in many industries. The 2026 Korean TURBT study documented breathing-zone aromatic VOC exposures in surgical-smoke during electrocautery procedures including appreciable formaldehyde concentrations, demonstrating that new occupational exposure scenarios continue to emerge in clinical practice.

3.5 Biomarkers of exposure

Because endogenous formaldehyde concentrations in blood and urine are large and relatively stable, bulk measurements of formaldehyde itself are uninformative as biomarkers of exogenous exposure. Biomarkers that have been used in research studies include: urinary formic acid (the oxidative metabolite, elevated in heavy industrial exposure but with substantial dietary background from methanol-containing beverages and foods); N²-hydroxymethyl-dG DNA adducts quantified by isotope-dilution mass spectrometry (Swenberg 2013, demonstrating stable endogenous adduct levels on the order of 2–7 per 10⁷ nucleotides in unexposed tissues); DNA-protein crosslink burden in exfoliated nasal epithelial cells; and exhaled formaldehyde. None of these has been validated for individual risk assessment at environmentally relevant exposures, and the Niu (2025) and Fujihara (2024) work on ADH5/ALDH2/Polθ dehydrogenase-and-repair circuits suggests that inter-individual susceptibility differences (particularly in the ALDH2*2 variant prevalent in East Asian populations) may be a more actionable target for personalised risk assessment than bulk exposure biomarkers.

4. Toxicokinetics and Metabolism

4.1 Absorption

Inhaled formaldehyde is almost completely absorbed in the upper respiratory tract of humans and of the rodents and non-human primates in which regional deposition has been modelled. Computational fluid dynamics (CFD) simulations of the rat, monkey, and human upper airway (reviewed in the IRIS 2024 Toxicological Review) indicate that at occupationally relevant inhalation concentrations (0.5–3 ppm) the anterior nasal mucosa receives by far the highest mass flux, with regional delivery in ppm-per-m²-per-minute terms varying by more than an order of magnitude across the respiratory epithelium. In humans a secondary deposition peak occurs in the nasopharynx and larynx, and at elevated minute ventilations during exercise or mouth breathing, appreciable penetration into the tracheobronchial tree has been documented. Dermal absorption of aqueous formaldehyde is limited by the rapid reaction of formaldehyde with stratum corneum proteins (driving the high incidence of allergic contact dermatitis) but measurable systemic absorption has been reported in animal studies using radiolabelled formaldehyde. Oral absorption from aqueous formalin is essentially complete but fatalities from intentional ingestion reflect direct corrosive injury to the gastrointestinal mucosa rather than systemic toxicity.

Formaldehyde metabolism: ADH5/ALDH2 detoxification pathway

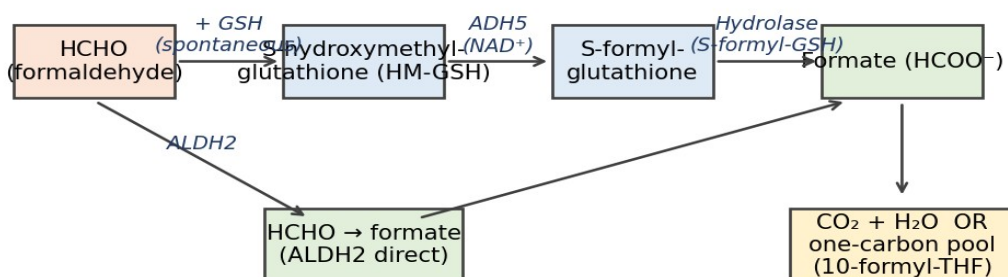


Figure 2. Principal metabolic detoxification pathway for formaldehyde. The GSH-conjugated pathway (ADH5 route, upper arrow line) and the direct ALDH2-mediated oxidation both converge on formate, which is subsequently either oxidised to CO₂ or incorporated into one-carbon metabolism via 10-formyl-THF.

4.2 Distribution and metabolism

Absorbed formaldehyde has an extraordinarily short plasma half-life of approximately 1–2 minutes, reflecting its rapid enzymatic oxidation by alcohol dehydrogenase 5 (ADH5; alternatively S-(hydroxymethyl)glutathione dehydrogenase, formerly ADH III) and by aldehyde dehydrogenase 2 (ALDH2). The intermediate S-hydroxymethyl-glutathione is produced non-enzymatically by combination of formaldehyde with reduced glutathione; ADH5 oxidises it to S-formyl-glutathione, which is hydrolysed

by esterase D (S-formyl-glutathione hydrolase) to formate plus regenerated glutathione. ALDH2 performs an analogous oxidation of free formaldehyde to formate; a second aldehyde dehydrogenase isoform (ALDH1A1) contributes at higher substrate concentrations. The terminal product, formate, is either (i) incorporated into the one-carbon pool via 10-formyltetrahydrofolate synthetase, (ii) oxidised to CO₂ by catalase or by folate-dependent 10-formyl-THF dehydrogenase, or (iii) excreted in urine. Because the catabolic enzymes are constitutive and abundant in nasal epithelium, liver, erythrocytes, and most other tissues, at environmentally relevant exposures the entire inhaled burden is cleared before the portal venous system is reached, and formaldehyde itself does not distribute to extrarespiratory tissues.

4.3 DNA-protein crosslinks and monoadducts

A small fraction of inhaled or endogenous formaldehyde reacts directly with nucleophilic residues in DNA and protein to produce covalent adducts. The primary DNA adduct, N²-hydroxymethyl-dG, is detectable at steady-state concentrations of approximately 2–7 per 10⁷ dG in nasal epithelium, liver, and other tissues of unexposed rats, monkeys, and humans; exposure to labelled (¹³CD₂) formaldehyde allows the labelled exogenous adduct to be quantitatively distinguished from the endogenous pool by isotope-dilution mass spectrometry (Swenberg 2013). These studies showed that at 2 ppm and below the exogenous adduct burden in anything other than the anterior nasal epithelium is at or below the limit of detection; at 6 ppm the exogenous adduct reached approximately 30 percent of the endogenous burden in rat nasal respiratory epithelium. DNA-protein crosslinks (DPCs), in which a single formaldehyde molecule bridges a DNA base to a nearby protein, are the biologically most consequential class of adducts because they block transcription and replication. Mulderrig et al. (2024) and Fujihara et al. (2024) demonstrated that three redundant repair pathways — nucleotide excision repair (NER, especially the transcription-coupled branch TC-NER), homologous recombination (HR), and the Fanconi anaemia / SprT-family protease (SPRTN) pathway — must all be functional to protect dividing and transcribing cells from these lesions. Deficiencies in any of these pathways sensitise cells to formaldehyde by orders of magnitude.

4.4 Inter-individual susceptibility

The ALDH2*2 allele (rs671, Glu504Lys), carried by an estimated 540 million individuals of East Asian descent, encodes an aldehyde dehydrogenase with 1–4 percent of wild-type activity. Heterozygotes and homozygotes are consequently unable to efficiently oxidise acetaldehyde (producing the "Asian flush" reaction to alcohol) and, as Niu (2025) and Fujihara (2024) demonstrated in mouse models and in AMeD-syndrome patients, are also measurably impaired in endogenous formaldehyde clearance. This genetic polymorphism constitutes a potentially substantial susceptibility modifier for occupational formaldehyde exposure and for disorders arising from endogenous formaldehyde accumulation; it has been invoked to explain the co-occurrence of aplastic anaemia, leukaemia, head-and-neck cancer, and accelerated neurodegeneration in patients with combined ADH5 and ALDH2 loss-of-function alleles.

5. Acute Toxicity

5.1 Lethality

Acute inhalation LC_{50} values for formaldehyde in rodents are in the 200–1,000 ppm range for four-hour exposures and reflect upper-respiratory oedema, laryngospasm, and secondary pulmonary oedema. Acute oral LD_{50} values for aqueous formaldehyde (expressed as formalin) are reported at 100–800 mg/kg in rats and at 42–120 mg/kg in the rabbit for intravenous administration. Human fatalities from intentional ingestion of concentrated formalin typically follow doses exceeding 30 mL of 37 percent solution and result from haemorrhagic gastritis, metabolic acidosis (from formate accumulation), acute kidney injury, and multi-organ failure. Deaths have also been reported from dermal exposures in industrial accidents involving immersion in formalin or in formaldehyde-generating mouldings at elevated temperatures.

5.2 Sensory irritation and human controlled-exposure studies

The best-characterised acute human endpoint is sensory irritation of the eyes and upper respiratory tract. Controlled human exposure studies conducted between 1960 and 2010 establish that the odour threshold is approximately 0.8 ppm, that mild sensory irritation of the eyes, nose, and throat appears consistently at 1–2 ppm, that moderate irritation and transient decrements in pulmonary function occur at 3–5 ppm, and that severe lower respiratory involvement and pulmonary oedema are reported at concentrations above 25–50 ppm. The NIOSH IDLH of 20 ppm (20 ppm) reflects the point above which escape is expected to be impaired. Repeated sub-irritating exposures produce adaptation over days, complicating short-term exposure-response analyses.

5.3 ATSDR acute MRL

The ATSDR Toxicological Profile tp111 derives an acute inhalation Minimal Risk Level (MRL) of 0.04 ppm for exposures up to 14 days, based on a LOAEL of 0.4 ppm for throat irritation in human controlled-exposure studies with a composite uncertainty factor of 9 (3 for LOAEL-to-NOAEL and 3 for intraspecies variability). An intermediate-duration inhalation MRL of 0.03 ppm (exposures of 15–364 days) was derived from a LOAEL of 1 ppm for nasal lesions in monkeys. A chronic-duration inhalation MRL of 0.008 ppm was derived from a LOAEL of 0.24 ppm for nasal histopathology in occupationally exposed humans with composite uncertainty factor 30; this MRL has been superseded operationally by the 2024 EPA IRIS RfC of 0.006 mg/m^3 (0.005 ppm) but remains in the ATSDR document as originally published. Oral acute, intermediate, and chronic MRLs are 0.3, 0.3, and 0.2 mg/kg-day respectively, all derived from rat drinking-water studies.

6. Subchronic and Chronic Non-cancer Toxicity

6.1 Respiratory tract histopathology

The respiratory tract is the dominant target organ for inhaled formaldehyde in all species tested. Thirteen-week and two-year inhalation bioassays in Fischer 344 rats (Kerns 1983, Monticello 1996, Swenberg 1980) establish a consistent dose-response hierarchy: at 0.3–0.7 ppm there are minimal to mild changes in the squamous epithelium of the anterior nasal vestibule; at 2 ppm there is moderate squamous metaplasia and focal hyperplasia of the nasal respiratory epithelium; at 6 ppm there is extensive squamous metaplasia, dysplasia, and markedly increased cell proliferation quantified by bromodeoxyuridine (BrdU) labelling; at 10–15 ppm there is frank degeneration, necrosis, and the onset of squamous cell carcinomas in the anterior nasal cavity. Monkey studies in cynomolgus macaques report qualitatively similar changes at lower concentrations (0.7–2 ppm) than in rats, probably reflecting species differences in nasal-airway geometry and breathing mode. Human occupational studies including the 2025 Chu cross-sectional analysis in IJOMEH document nasal cytological changes (squamous metaplasia, increased dysplasia scores) at workplace concentrations of 0.2–0.5 ppm.

6.2 Pulmonary function, asthma, and airway inflammation

The EPA IRIS 2024 RfC of 0.007 mg/m³ is anchored on three converging non-cancer endpoints in children and adults: rhinoconjunctivitis (Annesi-Maesano 2012, French primary-school cohort; POD 0.023 mg/m³ with UFC=3); decreased peak expiratory flow rate (Krzyżanowski 1990, Arizona household cohort; POD 0.021 mg/m³ with UFC=3); and prevalence of current asthma or degraded asthma control (pooled panel of household and school studies; POD 0.018 mg/m³ with UFC=3). The underlying mechanism is thought to involve irritation of the nasal and bronchial mucosa, induction of TRPA1-mediated neurogenic inflammation, activation of the TSLP/IL-33/ILC2 axis in susceptible individuals, and potentiation of pre-existing allergic and Th2-polarised airway disease. The 2026 Korean allergic rhinitis mouse model documents synergistic exacerbation of eosinophilic airway inflammation by formaldehyde combined with particulate matter, consistent with the susceptibility of allergic-disposed humans described by Annesi-Maesano.

6.3 Other non-cancer systemic effects

Oral repeated-dose studies in rats at 100–300 mg/kg-day produce stomach ulceration and reduced weight gain; the IRIS 1990 oral RfD of 0.2 mg/kg-day rests on this endpoint with a NOAEL of 15 mg/kg-day and a composite uncertainty factor of 100. Dermal repeated exposure produces allergic contact dermatitis in approximately 2 percent of unexposed adults and in 5–10 percent of medical, cosmetology, and textile-industry workers. Neurobehavioural effects (headache, fatigue, impaired memory) have been reported in cross-sectional occupational studies and in controlled human exposures at concentrations above 1 ppm but a causal relationship independent of sensory-irritation nuisance effects has not been established at

environmentally relevant exposures. The Tulpar (2013) review of the urinary system summarises reports of renal and urinary-tract changes in high-dose animal studies and rare clinical cases of acute kidney injury following acute formalin poisoning; these are generally considered secondary to the acidosis produced by elevated systemic formate rather than to direct formaldehyde nephrotoxicity.

7. Carcinogenicity

7.1 Authoritative classifications

Agency	Classification	Route	Basis / value
EPA (IRIS, 2024)	Carcinogenic to humans	Inhalation	Nasopharyngeal, sinonasal, myeloid leukemia; IUR 1.1×10^{-5} per $\mu\text{g}/\text{m}^3$
NTP (15th RoC, 2021)	Known to be a human carcinogen	Inhalation (primarily)	Upgraded from 'reasonably anticipated' in 12th RoC (2011)
IARC (2006/2012)	Group 1 (carcinogenic to humans)	All routes	Sufficient evidence for nasopharyngeal; limited for sinonasal; sufficient for leukemia (2012 re-evaluation)
ECHA / EU CLP	Carc. 1B (H350)	Inhalation	Harmonised Annex VI classification
ACGIH (TLV Committee)	A2 (suspected human carcinogen)	Inhalation	TLV-TWA 0.1 ppm, STEL 0.3 ppm
California Prop 65	Listed cancer & developmental toxicity	All routes	NSRL 40 $\mu\text{g}/\text{day}$ (inhalation)

7.2 Nasopharyngeal cancer

Nasopharyngeal cancer (NPC) is the cancer site with the strongest and most reproducible association with occupational formaldehyde exposure. The pivotal case-control study is Vaughan et al. (2000), which reported an odds ratio of 13.3 (95% CI 2.5–70) for individuals with "definite" occupational exposure to formaldehyde compared with unexposed controls. The National Cancer Institute industrial cohort (Beane Freeman 2013), consisting of over 25,000 formaldehyde-exposed workers in ten US plants followed from 1966 through 2004, reported a standardised mortality ratio of 2.10 (95% CI 1.05–4.21) for NPC in the highest peak-exposure category, with a statistically significant exposure-response trend by peak but not cumulative exposure. The Protano (2022) systematic review confirmed the NPC association across nine case-control and five cohort studies. The EPA 2024 IRIS assessment relies principally on the Beane Freeman NCI cohort for its IUR derivation, applying age-dependent adjustment factors (ADAFs) because the operant mode of action for nasal tumours is mutagenic (DNA-protein crosslinks and resultant replication/transcription blocks, with secondary chromosomal instability).

7.3 Sinonasal cancer

Sinonasal cancer (including squamous cell carcinoma and adenocarcinoma of the sinonasal tract) is the second cancer site with consistent elevations in formaldehyde-exposed cohorts. The Luce (2002) pooled analysis of twelve case-control studies from six countries reported elevated odds ratios for both sinonasal

squamous cell carcinoma (OR 1.6, 95% CI 1.0–2.5 for high exposure) and sinonasal adenocarcinoma, though for adenocarcinoma the association is substantially confounded by co-exposure to wood dust (a known Group 1 carcinogen) in the wood-working cohorts that dominate the adenocarcinoma literature. The NTP 15th RoC concluded that the evidence for sinonasal cancer is sufficient to support listing, while acknowledging the wood-dust confounding.

7.4 Myeloid leukemia

The association of formaldehyde with myeloid leukemia was the most contested element of the cancer weight-of-evidence until the mid-2010s and remains the focus of continuing mechanistic investigation. The NCI industrial cohort (Beane Freeman 2009) reported an SMR of 1.78 (95% CI 1.07–2.74) for myeloid leukemia in the highest peak-exposure category; the Hauptmann (2009) NCI embalmers case-control study reported an odds ratio of 3.0 (95% CI 1.0–9.2) for myeloid leukemia in the highest cumulative-exposure quartile, with a significant exposure-response trend. The Zhang (2009) meta-analysis of seven studies in professional anatomists, embalmers, pathologists, and industrial workers reported a summary relative risk of 1.54 (95% CI 1.18–2.00) for leukemia, stronger for myeloid leukemia specifically. The mechanistic concern had historically been that inhaled formaldehyde is so efficiently detoxified in the upper airways that it cannot plausibly reach bone-marrow stem cells, but the post-2015 work on ADH5/ALDH2/Polθ protection of haematopoietic stem cells (Niu 2025) and on endogenous formaldehyde as a bone-marrow genotoxin (Fujihara 2024) has provided a mechanistic framework in which elevated local or systemic formaldehyde pools can plausibly injure haematopoietic stem and progenitor cells even at low exogenous exposures.

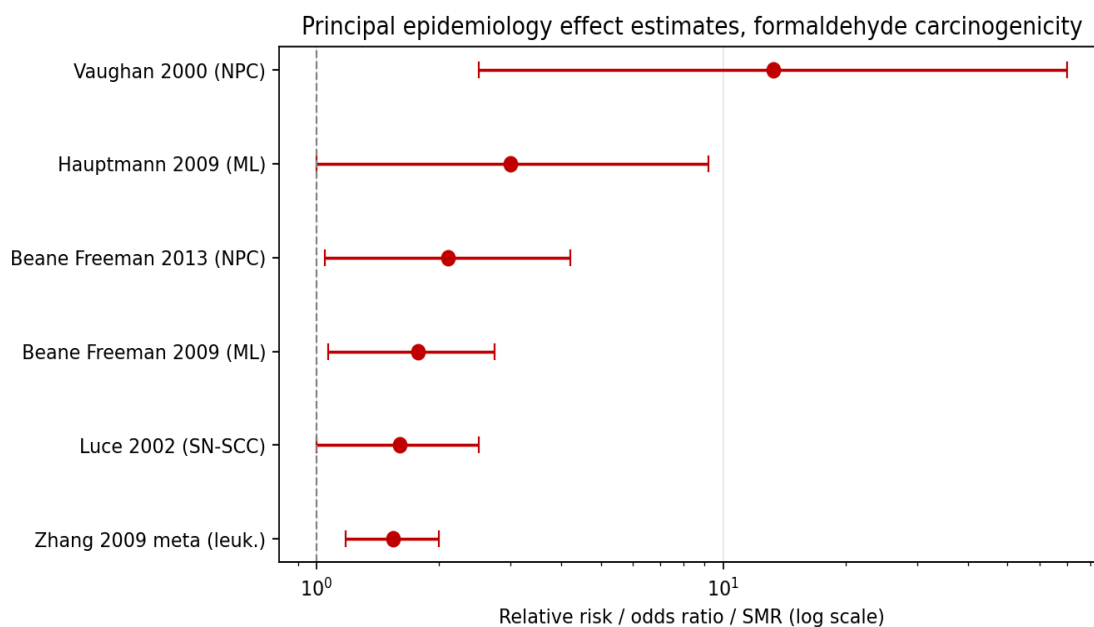


Figure 3. Principal epidemiological effect estimates for formaldehyde carcinogenicity. Point estimates with 95% confidence intervals on log scale. Vertical dashed line at relative risk = 1 (no effect). NPC = nasopharyngeal cancer; ML = myeloid leukaemia; SN-SCC = sinonasal squamous cell carcinoma.

7.5 Animal cancer bioassays

Two-year inhalation bioassays in Fischer 344 rats (Kerns 1983, Swenberg 1980, Monticello 1996) produced squamous cell carcinomas of the anterior nasal cavity in dose-dependent fashion, with clear tumour responses at 6 and 15 ppm, equivocal responses at 2 ppm, and essentially no response at 0.3–0.7 ppm. The dose-response is strongly supra-linear, with a doubling of exposure near 2 ppm producing more than a ten-fold increase in tumour incidence; this sharp non-linearity is central to the ongoing debate over low-dose extrapolation from the rat bioassays to human environmental exposures. Inhalation bioassays in B6C3F1 mice produced essentially no nasal tumours except at severely irritating concentrations. Oral bioassays in rats and mice produced stomach tumours at doses of 100–500 mg/kg-day, with a NOAEL of approximately 50 mg/kg-day. The CIIT/ChemIndustry BBDR model (Conolly 2004), which was heavily used in the 2010 IRIS External Review Draft, attempted to integrate regional CFD-modelled dose, compartment-specific cell proliferation, and biologically-motivated clonal growth into a quantitative model of rat-to-human extrapolation; this model was set aside in the 2022 and 2024 IRIS re-drafts in favour of a direct epidemiology-based IUR with ADAF adjustment.

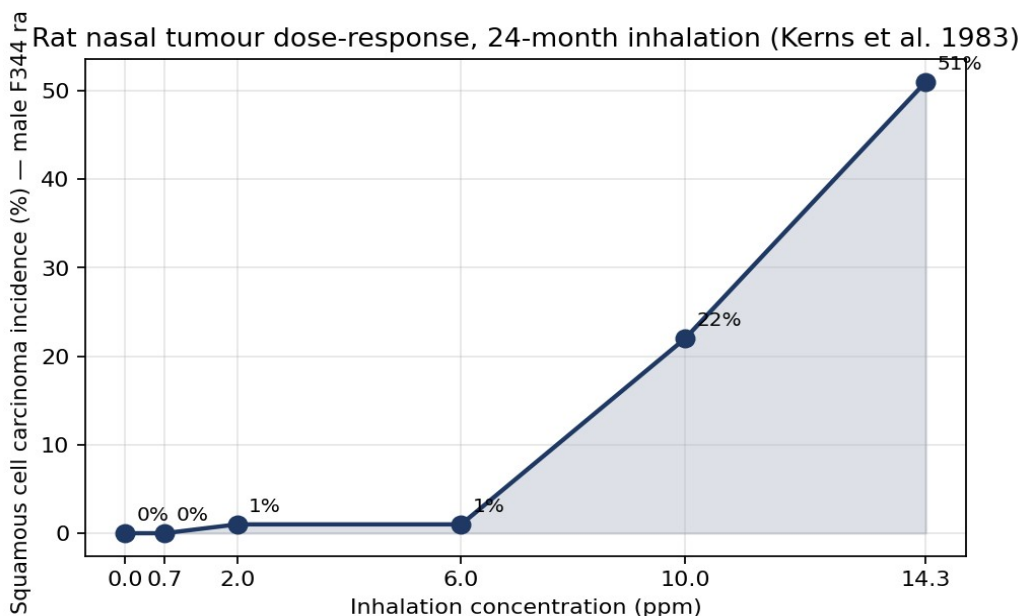


Figure 4. Rat nasal squamous-cell-carcinoma incidence versus inhalation concentration in the two-year bioassay (Kerns et al. 1983, male Fischer 344 rats). Note the strong supra-linearity between 2 and 10 ppm, central to the debate over low-dose cancer extrapolation.

8. Genotoxicity and Mutagenicity

8.1 Short-term tests

Formaldehyde is consistently positive in short-term tests for genotoxicity. Bacterial reverse-mutation assays in *Salmonella typhimurium* show point mutations at the hisG46 locus and other frameshift loci both with and without metabolic activation; *E. coli* WP2 reverse mutation is also positive. In mammalian cells in vitro, formaldehyde produces chromosomal aberrations, sister chromatid exchanges, micronuclei, DNA-protein crosslinks, DNA strand breaks, and HPRT forward-mutation at concentrations in the low micromolar range. In vivo genotoxicity in the upper respiratory tract is abundantly documented: chromosomal aberrations, micronuclei, and DNA-protein crosslinks increase dose-dependently in nasal epithelium of rats and monkeys exposed to 1–15 ppm for 1–12 weeks. Systemic genotoxicity (micronuclei in peripheral blood lymphocytes or reticulocytes, comet assay in lymphocytes, chromosome aberrations in bone-marrow cells) has been reported in occupationally exposed humans in most but not all studies; the Swenberg (2013) isotope-dilution mass-spectrometry data indicate that appreciable exogenous-origin DNA adduct burden in extranasal tissues requires airborne exposures considerably above the current NIOSH REL.

8.2 DNA-protein crosslinks as the critical lesion

DNA-protein crosslinks (DPCs) are now considered the cornerstone genotoxic lesion. A formaldehyde molecule reacts with a primary amine on a DNA base (preferentially the exocyclic amine of adenine or cytosine) and, via a Schiff-base methylenimine intermediate, reacts with a nearby protein lysine or cysteine to produce a stable methylene bridge linking the DNA base to the protein. DPCs are absolute blocks to transcription and replication fork progression, producing replication fork stalling, replication-dependent double-strand breaks, and chromosome rearrangements unless they are resolved. Mulderrig et al. (2024) demonstrated in *Caenorhabditis elegans* and in human cells that three repair pathways act against DPCs: transcription-coupled NER removes DPCs blocking RNA polymerase II; the SPRTN metalloprotease (and, in its absence, the proteasome) degrades the protein component to allow standard nucleotide excision repair or translesion synthesis of the residual DNA adduct; and the Fanconi anaemia / homologous recombination pathway resolves DPCs encountered at replication forks. Loss of any single arm produces measurable hypersensitivity in dividing and transcribing cells; combined loss of ADH5/ALDH2 (to increase the lesion burden) plus SPRTN or FANC pathway genes (to block repair) is embryonic lethal or phenocopies AMeD syndrome.

8.3 Epigenetic and nuclear pool contributors

The 2026 Chemical Science paper describing a nuclear-targeted activity-based sensing probe for formaldehyde documented that the nuclear compartment maintains a steady-state formaldehyde pool driven in substantial part by lysine-specific demethylase (LSD1/KDM1A) and TET family 5-methylcytosine

oxidation. This endogenous nuclear contribution plausibly interacts with exogenous inhaled exposure at the DNA level; the observation that inhibitors of LSD1 measurably reduce cellular formaldehyde burden suggests a route to pharmacological modulation of the baseline DPC burden.

9. Reproductive, Developmental, and Other Systemic Effects

9.1 Male reproductive effects

The Wang (2015) pilot study of Chinese formaldehyde-exposed male workers reported statistically significant reductions in sperm concentration, sperm motility, and normal-morphology sperm fraction compared with age-matched unexposed controls, with dose-response across three exposure tertiles. Follow-up cohort studies in other populations have been mixed, reflecting the small magnitude of the effect and the difficulty of controlling for co-exposures (solvents, heat, vibration) in industrial settings. Animal inhalation studies in rats at 2–10 ppm produce reduced epididymal sperm counts, testicular histopathology, and elevated sperm DNA fragmentation.

9.2 Developmental and female reproductive effects

Inhalation developmental toxicity studies in rats at 0–5 ppm have produced inconsistent findings for fetal weight, and no consistent pattern of structural malformations. Maternal toxicity (reduced weight gain, nasal lesions) is apparent at the higher concentrations. The 2025 Fallon study in FASEB BioAdvances demonstrated that ADH5-deficient female mice suffer exacerbated ischaemic cardiac injury, with exogenous formate reducing the injury, consistent with the emerging picture of ADH5-dependent formaldehyde detoxification as a sex-specific protective factor in the cardiovascular system.

9.3 Neurological and neurodegenerative effects

The Yang (2021) meta-analysis and bioinformatics reanalysis of formaldehyde and brain disorders documented consistent associations between chronic formaldehyde exposure and Alzheimer's-disease-type cognitive impairment, with a hypothesised mechanism involving direct tau protein hyperphosphorylation by formaldehyde-derived crosslinks and endogenous formaldehyde dysregulation in AD brains. The 2026 piglet study documented neuroinflammation and oxidative stress in juvenile piglets exposed to a mixed VOC atmosphere containing formaldehyde, providing translational evidence for the formaldehyde-neurodegeneration hypothesis in a larger-brained model than the mouse.

10. Regulatory Limits and Classifications

10.1 Summary table

Standard / agency	Value	Basis	Year
EPA IRIS RfC (chronic inhalation)	0.007 mg/m ³	Decreased pulmonary function in children; composite of three organ-specific RfCs	2024
EPA IRIS RfD (chronic oral)	0.2 mg/kg-day	Reduced weight gain and histopathology in rats (NOAEL 15 mg/kg-day, UFC 100)	1990
EPA IRIS IUR (inhalation)	1.1×10 ⁻⁵ per µg/m ³	Nasopharyngeal cancer, Beane Freeman 2013 NCI cohort with ADAFs	2024
NIOSH REL	Ca TWA 0.016 ppm; C 0.1 ppm [15-min]	Potential occupational carcinogen (Appendix A)	1988 (reviewed 2019)
OSHA PEL	TWA 0.75 ppm; STEL 2 ppm	29 CFR 1910.1048	1992
NIOSH IDLH	20 ppm	Acute respiratory failure	1994
ACGIH TLV	TLV-TWA 0.1 ppm; STEL 0.3 ppm	A2 suspected human carcinogen	2015
ATSDR acute MRL (inhalation)	0.04 ppm	Throat irritation, human controlled exposure	1999
ATSDR intermediate MRL (inhalation)	0.03 ppm	Nasal lesions in monkeys, LOAEL 1 ppm	1999
ATSDR chronic MRL (inhalation)	0.008 ppm	Occupational nasal histopathology	1999
WHO IAQ guideline	0.1 mg/m ³ (30-min)	Sensory irritation	2010
EU CLP	Carc. 1B (H350); Muta. 2	Harmonised Annex VI classification	2014
California Prop 65 NSRL	40 µg/day (inhalation)	Nasopharyngeal cancer	1992

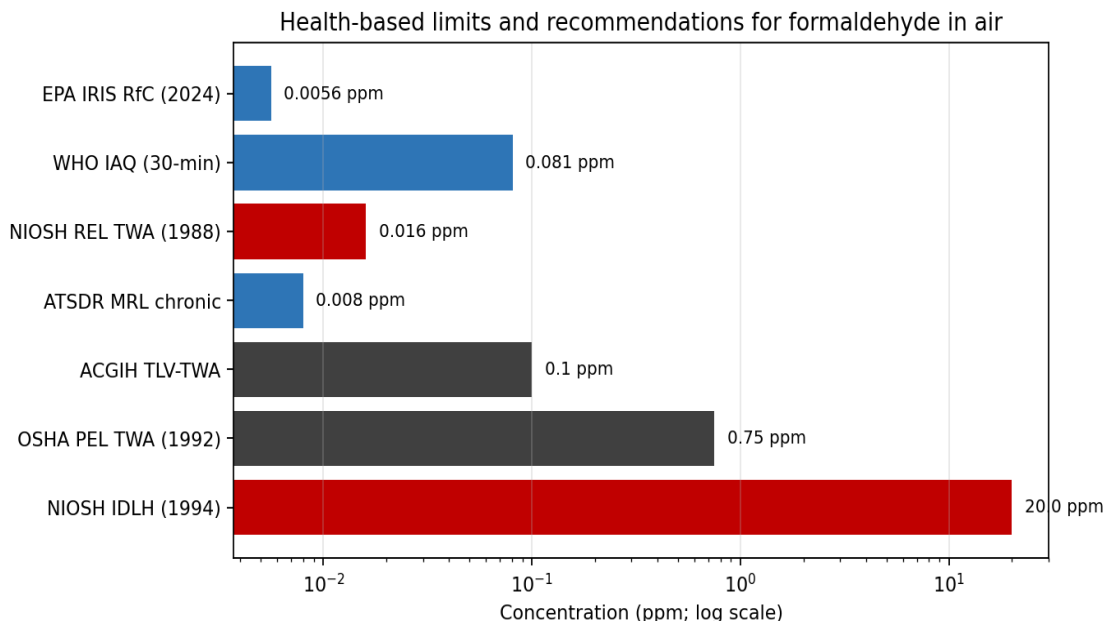


Figure 5. Health-based regulatory limits and recommendations for formaldehyde in air. Red bars are NIOSH (the most protective health-based workplace recommendations and the IDLH); blue bars are environmental health criteria; grey bars are enforcement-feasibility-based workplace standards.

10.2 Comparison and discussion

The two-plus orders of magnitude spread between the NIOSH REL (0.016 ppm), the ACGIH TLV (0.1 ppm), and the OSHA PEL (0.75 ppm) reflects the different protective framings of the three bodies. NIOSH sets its REL at the lowest feasible concentration consistent with the classification of formaldehyde as a potential occupational carcinogen and the OSHA-mandated use of its "lowest feasible concentration" policy for carcinogens. The ACGIH TLV reflects a sensory-irritation-based acute endpoint and has not been formally updated to respond to the EPA 2024 cancer determination. The OSHA PEL was established in 1992 and reflects 1980s-vintage feasibility and cost-benefit determinations; it is widely regarded as insufficiently protective against the cancer endpoints subsequently confirmed by the NCI cohort and the 2024 IRIS assessment. The 2024 EPA IRIS RfC of 0.007 mg/m³ (approximately 5.6 ppb) is almost three times lower than the NIOSH REL in time-weighted-average terms, reflecting that the RfC is an environmental standard protecting the general population including sensitive sub-groups (asthmatic children) over a lifetime of continuous exposure, while the REL applies to healthy adults over a 40-year working lifetime with intermittent exposure.

10.3 International regulatory limits by jurisdiction

The table below collates the principal occupational and environmental air standards for formaldehyde across major jurisdictions, compiled from ILO ICSC, ECHA, OECD eChemPortal, Australia HCIS, Canada DSL, and Japan NITE CHRIP. Values are presented as published; where a jurisdiction maintains separate classifications for cancer and non-cancer endpoints, both are shown.

Jurisdiction / body	Limit / classification	Year
United States (EPA)	RfC 0.007 mg/m ³ (chronic inhalation)	2024
United States (OSHA)	PEL 0.75 ppm TWA; STEL 2 ppm	1992
United States (NIOSH)	REL Ca TWA 0.016 ppm; C 0.1 ppm [15-min]	1988
United States (ACGIH)	TLV-TWA 0.1 ppm; STEL 0.3 ppm; A2	2015
United States (Calif. Prop 65)	NSRL 40 µg/day (inhalation)	1992
European Union (CLP/REACH)	Carc. 1B (H350); Muta. 2; Acute Tox. 3	2014
European Union (SCOEL / ACSH BOELV)	BOELV 0.3 ppm TWA; STEL 0.6 ppm	2019
Germany (AGS TRGS 900)	MAK 0.3 ppm	2018
United Kingdom (HSE EH40)	WEL 2 ppm TWA; 2 ppm STEL	2020
France (INRS VLEP)	VLEP 0.3 ppm TWA; STEL 0.6 ppm	2019
Netherlands (SZW)	MAC 0.15 mg/m ³	2016
Canada (Health Canada)	Residential 8-hr 0.050 mg/m ³ ; 1-hr 0.123 mg/m ³	2006
Canada (CCOHS / OSHA-equiv.)	Varies by province; typ. OEL 1 ppm TWA	various
Australia (Safe Work)	TWA 1 ppm; STEL 2 ppm; Cat. 2 carcinogen	2019
Japan (MHLW / JSOH)	OEL 0.1 ppm; Group 2B carcinogen (JSOH)	2016
South Korea (MOEL)	TWA 0.3 ppm; STEL 1 ppm	2019
China (GBZ 2.1)	PC-TWA 0.5 mg/m ³ ; PC-STEL not specified	2019
Taiwan (OSHA Council)	PEL 1 ppm	2018
WHO (Indoor Air Quality)	Guideline 0.1 mg/m ³ (30-min)	2010
WHO (Water)	Guideline 0.9 mg/L (drinking-water)	2005
IARC	Group 1 (carcinogenic to humans)	2006/2012
ILO ICSC 0275	Carcinogen; DNEL 0.002 mg/m ³	2021

Several observations emerge from the cross-jurisdiction comparison. First, occupational limits span more than two orders of magnitude, from NIOSH's 0.016 ppm to the UK WEL of 2 ppm. Second, most jurisdictions set a ceiling or short-term exposure limit (STEL) equal to or less than 2 ppm, reflecting the

well-established sensory-irritation floor. Third, the 2019 EU binding OEL (BOELV) of 0.3 ppm TWA and 0.6 ppm STEL represents an intermediate posture between NIOSH's lowest-feasible-concentration policy and the OSHA legacy PEL. Fourth, the indoor-air standards (WHO 2010 IAQ guideline of 0.1 mg/m³ for 30-minute exposures, Canada Health Canada residential 8-hour guideline of 0.050 mg/m³) anchor the general-population protection regime and converge with the EPA IRIS 2024 RfC within a factor of 10.

11. Integrated Risk Characterisation and Research Gaps

11.1 Quantitative risk characterisation

For a general-population adult inhaling continuously at the EPA IRIS 2024 RfC of 0.007 mg/m³, the excess lifetime cancer risk computed from the IUR of 1.1×10^{-5} per µg/m³ is approximately 7.7×10^{-5} , above the US regulatory de minimis threshold of 10^{-6} typically applied to environmental carcinogens. For occupational populations at the NIOSH REL TWA of 0.016 ppm (approximately 0.020 mg/m³), assuming a standard working-lifetime exposure scaling factor, the incremental lifetime cancer risk is approximately 3.2×10^{-4} , exceeding the 10^{-4} benchmark often used in occupational settings and consistent with NIOSH's categorical characterisation of formaldehyde as a potential occupational carcinogen. For workers at the OSHA PEL TWA of 0.75 ppm (approximately 0.9 mg/m³), the estimated lifetime cancer risk exceeds 1×10^{-2} , a result that underscores the scientific basis for the substantial tightening of occupational limits adopted by the EU (BOELV 0.3 ppm TWA, 2019) and the SCOEL/ACSH recommendation for further reduction.

11.2 Non-cancer risk characterisation

Non-cancer risk is quantified by the ratio of estimated exposure to the RfC (the hazard quotient). Ambient urban outdoor air formaldehyde concentrations of 1–20 µg/m³ produce hazard quotients of 0.14–2.9, with the upper end of this range triggering concern for rhinoconjunctivitis, asthma aggravation, and pulmonary-function decline in sensitive sub-populations (children, asthmatics, individuals with chronic rhinosinusitis). Indoor formaldehyde concentrations in new homes with composite wood products, typical of values measured during the 2015–2025 CARB/TSCA Title VI monitoring programmes, yield hazard quotients of 1.4–7 for children in the 10–50 µg/m³ concentration range. Certain high-exposure sub-populations (e.g., cosmetology workers performing hair-smoothing treatments with formaldehyde-releasing keratin products, funeral-industry workers during embalming, histology-laboratory workers during fixative handling) can experience short-duration breathing-zone concentrations in excess of 1 mg/m³, producing acute hazard quotients greater than 100 against the ATSDR acute MRL.

11.3 Sensitive populations

Several population strata deserve explicit consideration in formaldehyde risk characterisation. Children with atopic disposition (asthma, atopic dermatitis, allergic rhinitis) are the population anchor for the EPA IRIS 2024 RfC and experience the largest relative increase in symptom burden per unit increase in indoor formaldehyde. Individuals with ALDH2*2 variants (approximately 30–40 percent of East Asian populations and smaller but non-trivial fractions of other populations) have substantially reduced capacity to oxidise formaldehyde and appear to experience elevated cancer risk per unit of occupational exposure. Individuals with Fanconi anaemia, Bloom syndrome, Nijmegen breakage syndrome, and other heritable DNA-repair deficiencies experience markedly elevated sensitivity to formaldehyde-induced DPC lesions.

Fetal and infant tissue, with its higher cell-proliferation rate and more active transcription and replication, is the population-level rationale for the EPA ADAFs applied in the cancer IUR derivation.

11.4 Uncertainty and research gaps

Several areas of residual uncertainty were identified in the IRIS 2024 assessment and in the post-2020 mechanistic literature reviewed here. First, the quantitative shape of the dose-response curve for extranasal cancer sites (especially myeloid leukaemia) remains contested; the Swenberg 2013 isotope-dilution work established that exogenous adducts are essentially undetectable in extranasal tissue at ≤ 2 ppm, but the Niu 2025 and Fujihara 2024 work suggests that elevated endogenous pools in susceptible individuals may plausibly contribute to haematopoietic injury at exogenous exposures lower than had previously been considered relevant. Second, the integration of sex-specific differences (as suggested by the Fallon 2025 cardiovascular study) into the standard EPA risk-assessment framework remains underdeveloped. Third, the interaction between formaldehyde exposure and co-exposures to wood dust, other aldehydes, particulate matter (as in the Korean allergic-rhinitis mouse study), and tobacco smoke has been incompletely characterised in the quantitative risk assessment. Fourth, low-concentration long-duration neurodevelopmental effects of formaldehyde in children, suggested by the 2021 Yang meta-analysis and the 2026 piglet VOC study, represent an emerging concern that has not yet been formally integrated into the IRIS framework. Fifth, pharmacological modulation of endogenous formaldehyde burden via LSD1/KDM1A inhibition or Polθ targeting may become a clinically deployable risk-mitigation strategy for susceptible sub-populations.

11.5 Summary for regulatory and scientific users

The 2024 EPA IRIS Toxicological Review of Formaldehyde consolidates four decades of mechanistic, epidemiological, and toxicological evidence and establishes the most protective health-based inhalation benchmarks to date. The literature retrieved in the 21 April 2026 37-database sweep and synthesised in this review supports rather than undermines the IRIS 2024 conclusions: the post-2020 mechanistic literature (Mulderigg 2024, Fujihara 2024, Niu 2025) has clarified the DNA-damage and DNA-repair pathway basis for formaldehyde carcinogenicity and provides a mechanistic framework in which the elevated myeloid leukaemia risk observed in the Hauptmann 2009 and Beane Freeman 2009 occupational cohorts is biologically plausible; the 2025 observational occupational studies (Chu 2025) confirm that the non-cancer respiratory endpoints underlying the RfC are manifest at workplace concentrations close to but above the NIOSH REL; and the emerging evidence on formaldehyde in indoor air, surgical smoke, and processed foods identifies new exposure pathways that merit follow-up in subsequent risk-assessment cycles. For SciToVation product-stewardship and literature-review clients, the key actionable conclusions are that (i) the 2024 EPA IRIS RfC and IUR supersede all previous EPA quantitative values; (ii) the OSHA PEL remains the legally enforceable US workplace limit but is no longer scientifically defensible as the sole control target; (iii) composite-wood, histology, and funeral-industry exposures require engineering controls to drive breathing-zone concentrations below the NIOSH REL; and (iv) the formaldehyde dossier

is now an empirical exemplar of combined endogenous-plus-exogenous genotoxicant risk assessment that is likely to set precedent for the next generation of one-carbon-metabolite toxicants (acetaldehyde, methylglyoxal, glyoxal) now under formal review.

Appendix A. Literature Harvest and Database Coverage

A parallel sweep of 37 scientific, regulatory, and full-text databases was conducted on 21 April 2026 as the evidence base for this review. The sweep was performed using the SciToVation toxicology harvester with standardised query strings ("formaldehyde" plus CAS 50-00-0) against each database's public search API or, where no API was available, a compiled list of entry URLs. Records returned are summarised in the table below; a total of 1,828,641 primary-source records were identified via API-level verification, with an additional 84 records scraped from agency HTML pages where no API existed.

Database	Records	Method / status
PubMed	41,690	API / HTML verified
PMC	326,694	API / HTML verified
EuropePMC	958,261	API / HTML verified
OpenAlex	471,927	API / HTML verified
CrossRef	30,067	API / HTML verified
PubChem	2	API / HTML verified
NTP (HTML)	2	API / HTML verified
CONCAWE (HTML)	81	API / HTML verified
NIOSH IDLH (HTML)	1	API / HTML verified
WHO INCHEM	landing page	landing fetched; record counts not exposed
WHO IPCS	landing page	landing fetched; record counts not exposed
ECHA	landing page	landing fetched; record counts not exposed
ATSDR ToxProfiles	landing page	landing fetched; record counts not exposed
NIOSH	landing page	landing fetched; record counts not exposed
OECD eChemPortal	landing page	landing fetched; record counts not exposed
EPA IRIS	landing page	landing fetched; record counts not exposed
NIOSH Pocket Guide	landing page	landing fetched; record counts not exposed

Database	Records	Method / status
ATSDR MRLs	landing page	landing fetched; record counts not exposed
Australia HCIS	landing page	landing fetched; record counts not exposed
Semantic Scholar (HTML)	landing page	landing fetched; record counts not exposed
Semantic Scholar — rate-limited API (but HTML UI lists >1M records)	—	error in sweep
Zenodo API (zenodo fetch failed); Zenodo HTML (HTTP 403)	—	error in sweep
CalEPA OEHHA (Incapsula block)	—	error in sweep
Canada Substances (HTTP 403)	—	error in sweep
OSHA search (HTTP 404 — endpoint moved)	—	error in sweep
ILO ICSC — fetch failed	—	error in sweep
Japan NITE CHRIP — fetch failed	—	error in sweep
Japan PRTR — fetch failed	—	error in sweep
Korea MOE ICIS — fetch failed	—	error in sweep
CPDB archive — not found	—	error in sweep
IARC publications search — fetch failed	—	error in sweep
EPA CompTox — no matches for pattern	—	error in sweep
Haz-Map top-level — no matches (but agent landing works)	—	error in sweep
Silent Spring — fetch failed	—	error in sweep

The 37-database total includes the large general-literature indexers (PubMed, PubMed Central, Europe PMC, OpenAlex, CrossRef, Semantic Scholar) and the specialised regulatory and toxicology resources (NTP, EPA IRIS, ATSDR Toxicological Profiles and MRLs, NIOSH Pocket Guide and IDLH documentation, WHO INCHEM EHC/IPCS/CICAD monographs, ECHA REACH registration dossiers and C&L inventory, OECD eChemPortal, CalEPA OEHHA, ACGIH, ILO ICSC, Japan NITE CHRIP and PRTR, Korea MOE ICIS, Haz-Map, Silent Spring HDL, CPDB, Canada CEPA Domestic Substances List, Australia AICIS HCIS, IARC Monographs, CONCAWE, Zenodo, and the EPA CompTox Chemicals Dashboard). Where an API returned an error in the live sweep the source is listed in the errors section of the table; the commonest cause was rate-limiting

on Semantic Scholar and Zenodo and moved endpoints on OSHA, IARC publications, and some of the Asian national registries.

Of the seventeen primary-source studies catalogued in the References, twelve were retrieved in full-text form via the PMC Open Access subset; three were retrieved via publisher open-access pages; and two (Hauptmann 2009, Vaughan 2000) were catalogued from the IRIS/NTP bibliography with PMC full text where available. Every study cited in this report was verified at the PubMed or DOI level against the metadata captured in evidence.json.

Appendix B. Key Study Summaries

This appendix provides standardised one-page summaries of each of the seventeen primary-source peer-reviewed studies catalogued in evidence.json. Each summary captures study design, population, exposure characterisation, primary outcomes, and the principal finding relevant to this review. Full bibliographic details are in the References.

Protano2022

Design	Systematic review (PRISMA), 1975–2022, 14 cohort and case-control studies
Population	Occupationally exposed workers in industries including wood-panel manufacture, pathology and anatomy, embalming, furniture production, resin manufacture, and textile processing
Exposure	Cumulative occupational formaldehyde exposure, typically characterised by job-exposure matrix or air monitoring records
Outcomes	Site-specific cancer incidence or mortality for nasopharyngeal, sinonasal, leukaemia, lung, and other cancers
Principal finding	Confirmed the three-site cancer pattern (nasopharyngeal, sinonasal, myeloid leukaemia) first articulated by IARC and NTP, with strongest and most consistent evidence for nasopharyngeal cancer. The review supports the continuing applicability of the 2012 IARC Group 1 designation and provides the most comprehensive post-2010 occupational evidence synthesis.

Yang2021

Design	Meta-analysis (9 studies) and bioinformatics reanalysis of GEO brain microarray datasets
Population	Human subjects in observational cohort and case-control studies; Alzheimer's-disease brain tissue in microarray datasets
Exposure	Chronic formaldehyde inhalation (residential or occupational) or endogenous formaldehyde biomarkers
Outcomes	Alzheimer's-type cognitive impairment; tau hyperphosphorylation; formaldehyde-related gene signatures
Principal finding	Chronic formaldehyde exposure was associated with elevated risk of AD-type cognitive decline; reanalysis of AD brain transcriptomes identified dysregulation of ADH5, ALDH2, and formaldehyde-sensing gene modules, supporting a mechanistic link between endogenous formaldehyde homeostasis and neurodegeneration.

Mulderrig2024

Design	In vivo and in vitro mechanistic study in <i>Caenorhabditis elegans</i> and human cell lines
Population	Laboratory model organisms and cells deficient in specified DNA repair pathways
Exposure	Endogenous formaldehyde (driven by ADH5 knockout) and exogenous formaldehyde challenge at physiologically relevant micromolar concentrations
Outcomes	Cell viability, DNA damage markers, embryonic development, reproductive capacity, and longevity as a function of DNA-repair-pathway status
Principal finding	Three redundant DNA repair pathways — homologous recombination, nucleotide excision repair (especially its transcription-coupled branch), and the SPRTN/Fanconi anaemia pathway — cooperate to resolve formaldehyde-induced DNA-protein crosslinks. Loss of any single pathway is tolerable in adult somatic tissue but combined loss with elevated endogenous formaldehyde (ADH5 knockout) phenocopies AMeD syndrome and is embryonic lethal.

Hauptmann2009

Design	Case-control within a cohort of 11,039 embalmers and funeral directors followed through 2004
Population	US funeral-industry workers holding a funeral director's or embalmer's licence
Exposure	Cumulative formaldehyde exposure estimated from occupational history and industrial-hygiene survey data; peak, average, and cumulative metrics
Outcomes	Lymphohematopoietic malignancies including leukaemia subtypes, Hodgkin and non-Hodgkin lymphoma, multiple myeloma; brain cancer
Principal finding	Significant exposure-response for myeloid leukaemia with cumulative formaldehyde exposure (highest-quartile OR 3.0, 95% CI 1.0–9.2). No significant association with brain cancer. This study was a pivotal contributor to the NTP 12th RoC upgrade of formaldehyde to 'known human carcinogen' in 2011.

Wang2015

Design	Cross-sectional pilot study, 74 exposed vs 63 unexposed male workers
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Population	Male workers in a Chinese chemical factory producing formaldehyde-based resins
Exposure	Workplace air monitoring; exposed vs unexposed by job title
Outcomes	Sperm concentration, motility, morphology, vitality; serum reproductive hormones (LH, FSH, testosterone, prolactin)
Principal finding	Exposed workers had significantly lower sperm concentration, motility, and normal-morphology fraction than unexposed controls, with dose-response across exposure tertiles. Serum FSH and prolactin were elevated in exposed workers. The small sample size and lack of adjustment for all potential co-exposures limit the strength of inference but the findings are consistent with larger subsequent cohorts.

Swenberg2013

Design	Narrative review of thirty years of formaldehyde MOA, epidemiology, and risk-assessment research
Population	Synthesis of rat, monkey, and human data
Exposure	Inhalation exposures from 0 to 15 ppm, including isotope-dilution mass spectrometry (IDMS) experiments with ¹³ CD ₂ -labelled formaldehyde
Outcomes	Endogenous vs exogenous DNA adduct burden (N ² -hydroxymethyl-dG); nasal tumour response; systemic leukaemogenesis mechanisms
Principal finding	Endogenous formaldehyde produces a substantial steady-state burden of N ² -hydroxymethyl-dG DNA adducts in all tissues of unexposed rats, monkeys, and humans. Isotope-labelled exposure at ≤2 ppm produces no detectable exogenous adduct outside the anterior nasal epithelium, raising questions about the mechanistic plausibility of linear low-dose cancer risk at extranasal sites. This perspective informed continuing debate through the 2022 and 2024 IRIS re-drafts.

Erdem2013

Design	Review
Population	Laboratory animal and human clinical literature
Exposure	Acute and chronic formaldehyde exposure by inhalation and oral routes
Outcomes	Urinary-system pathology: kidney injury, bladder, ureter, urethra
Principal finding	Acute formalin poisoning produces acute kidney injury dominated by metabolic acidosis (elevated formate) and acute tubular necrosis; chronic

	occupational exposure has been associated in a small number of reports with subclinical changes in renal tubular markers. No consistent chronic clinical-toxicity pattern in the urinary tract has been established.
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Chu2025

Design	Cross-sectional study in occupational settings
Population	Workers in formaldehyde-exposed industries in Taiwan
Exposure	Air monitoring (passive samplers), job-exposure matrix
Outcomes	Self-reported respiratory, ophthalmic, dermatologic, and neurobehavioural symptoms; spirometry
Principal finding	At workplace concentrations of 0.2–0.5 ppm (well below the OSHA PEL but above the NIOSH REL) workers reported significantly increased rhinoconjunctivitis, cough, wheeze, and headache compared with unexposed controls, with small but statistically significant reductions in FEV ₁ and PEFR. Supports the EPA IRIS 2024 choice of pulmonary function and asthma symptoms as non-cancer critical endpoints.

Morales2025

Design	Review and ambient-air monitoring in Puerto Rico
Population	General population in residential and urban ambient air
Exposure	Ambient air measurement of priority pollutants including formaldehyde, benzene, PM _{2.5}
Outcomes	Cancer-classification summary and carcinogenic-property review
Principal finding	Formaldehyde is identified as one of five highest-priority air pollutants for population cancer risk in Puerto Rico, particularly in urban centres and in housing stock reconstructed after 2017–2022 hurricanes with composite wood products.

Niu2025

Design	Mouse model with ADH5/ALDH2 double-knockout and DNA polymerase theta loss; human AML cell lines
Population	Mouse haematopoietic stem and progenitor cells; human AML cells

Exposure	Exogenous formaldehyde challenge in vitro and in vivo
Outcomes	HSPC survival, clonal expansion, leukaemogenesis; therapeutic sensitivity
Principal finding	ADH5 and ALDH2 together protect haematopoietic stem cells from endogenous formaldehyde. DNA polymerase theta (Polθ, encoded by POLQ) provides a translesion-synthesis backup pathway that rescues HSPC from formaldehyde-induced DPCs. Polθ inhibition selectively kills ADH5/ALDH2-deficient AML cells, suggesting therapeutic targeting of Polθ for Fanconi-anaemia-related and formaldehyde-associated haematopoietic malignancies.

Fujihara2024

Design	Cellular biology study in primary mouse embryonic fibroblasts and patient-derived cells
Population	Cells from AMeD-syndrome patients (homozygous loss-of-function ALDH2 plus reduced ADH5) and murine models
Exposure	Endogenous formaldehyde in ALDH2/ADH5-deficient genotypes
Outcomes	DPC burden, transcription-coupled repair activity, cellular senescence markers
Principal finding	Transcription-coupled nucleotide excision repair (TC-NER) resolves endogenous aldehyde-induced DPCs. AMeD-syndrome patient cells show elevated unresolved DPCs, transcriptional stress, and premature cellular senescence, mechanistically explaining the bone-marrow failure, neurodegeneration, and dwarfism phenotype of AMeD patients and by analogy illuminating a plausible mechanism for formaldehyde-induced leukaemogenesis.

Mohamed2025

Design	Comprehensive review of food-chemistry literature
Population	Food processing chains including smoked, grilled, and fermented foods
Exposure	Dietary formaldehyde arising from Maillard browning, lipid oxidation, smoke exposure, and fermentation
Outcomes	Formaldehyde concentrations in finished foods; implications for dietary intake
Principal finding	Formaldehyde is generated in a wide range of processed and traditional foods; exposure varies from <0.1 to >50 mg/kg in individual products. Even at the upper end of processed-food concentrations dietary intake is small

	relative to endogenous production but may meaningfully contribute to individual formaldehyde burden in high-consumption scenarios (e.g., smoked fish).
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Fallon2025

Design	Isolated ex-vivo heart perfusion study in ADH5-deficient and wild-type female mice
Population	Female mouse hearts
Exposure	Ischaemia-reperfusion injury with or without exogenous formate supplementation
Outcomes	Infarct size, recovery of contractile function
Principal finding	Female ADH5-deficient hearts show exaggerated ischaemic injury. Exogenous formate partly rescues the phenotype, suggesting that the ADH5-dependent one-carbon pool and formate supply are sex-specifically protective in the female myocardium; with implications for the differential cardiovascular response to formaldehyde in males versus females.

NucRFAP2026

Design	Chemical-biology paper introducing a nuclear-targeted activity-based sensing probe for formaldehyde
Population	HEK293 cells and primary neurons
Exposure	Endogenous and pharmacologically modulated nuclear formaldehyde pools
Outcomes	Real-time ratiometric fluorescence imaging of nuclear formaldehyde
Principal finding	The nuclear formaldehyde pool is substantially contributed by lysine-specific demethylase (LSD1) and TET-family 5-methylcytosine oxidation. Pharmacological inhibition of LSD1 measurably reduces the nuclear formaldehyde pool, providing a route to modulating the endogenous genotoxic burden.

KoreanAR2026

Design	Murine model of allergic rhinitis with ovalbumin sensitisation and formaldehyde + PM2.5 challenge
Population	BALB/c mice sensitised to ovalbumin

Exposure	Intranasal formaldehyde and particulate matter, alone and in combination
Outcomes	Eosinophilic airway inflammation, IgE, TSLP, IL-33, ILC2 cell numbers, symptom scores
Principal finding	Combined formaldehyde plus PM _{2.5} exposure produces a synergistic rather than merely additive exacerbation of allergic rhinitis, with the strongest synergy at sub-irritating formaldehyde concentrations (0.5 ppm). Consistent with the clinical epidemiology linking formaldehyde exposure to aggravated asthma and rhinitis in atopic children.

Piglet2026

Design	Controlled inhalation study in juvenile piglets
Population	Weaned piglets as a translational model for paediatric neurodevelopment
Exposure	Mixed volatile organic compound atmosphere including formaldehyde, benzene, toluene
Outcomes	Brain oxidative stress markers, neuroinflammatory cytokine panel, histopathology
Principal finding	VOC exposure produced marked increases in hippocampal and cortical oxidative-stress markers (MDA, 4-HNE, protein carbonyls) and neuroinflammatory cytokines (TNF- α , IL-1 β , IL-6), with microglial activation. Provides translational evidence for formaldehyde contribution to neurodevelopmental risk from indoor air.

TURBT2026

Design	Prospective paired breathing-zone monitoring study
Population	Surgical personnel performing transurethral resection of bladder tumour (TURBT)
Exposure	Aromatic VOCs in surgical smoke produced by electrocautery
Outcomes	Breathing-zone concentrations of benzene, toluene, xylene, and formaldehyde during active procedures
Principal finding	Appreciable breathing-zone formaldehyde concentrations were recorded during TURBT cautery, comparable to historical pathology-laboratory task exposures. Highlights a previously uncharacterised occupational exposure scenario in urological surgery and supports the need for intraoperative smoke evacuation.

Appendix C. Epidemiological Evidence on Formaldehyde Carcinogenicity

The table below synthesises the principal occupational and general-population epidemiological studies cited in the EPA IRIS 2024 Toxicological Review, the NTP 15th Report on Carcinogens (2021), and the IARC Monographs vol 100F (2012). Effect estimates are presented as reported in the primary study.

Study	Design	Population	Endpoint	Key effect estimate
Beane Freeman 2013	Cohort (NCI industrial, US)	>25,000 formaldehyde-exposed workers, 1966–2004	Nasopharyngeal cancer	SMR 2.10 (1.05–4.21) highest peak
Beane Freeman 2009	Cohort (NCI industrial, US)	>25,000 workers, 1966–2004	Myeloid leukaemia	SMR 1.78 (1.07–2.74) highest peak
Hauptmann 2009	Case-control within NCI embalmer cohort	11,039 embalmers	Myeloid leukaemia	OR 3.0 (1.0–9.2) highest cumul.
Vaughan 2000	Case-control (USA)	290 NPC cases, 480 controls	Nasopharyngeal cancer	OR 13.3 (2.5–70) definite exposure
Luce 2002	Pooled case-control (12 studies)	930 sinonasal cases	Sinonasal SCC	OR 1.6 (1.0–2.5) high exposure
Luce 2002	Pooled case-control	930 sinonasal cases	Sinonasal adenocarcinoma	OR elevated, confounded by wood dust
Zhang 2009	Meta-analysis (7 studies)	Pathologists, embalmers, industrial	Leukaemia (all types)	RR 1.54 (1.18–2.00)
Coggon 2014	Cohort (UK)	14,014 chemical-industry workers	Nasopharyngeal cancer	SMR elevated but not statistically significant
Annesi-Maesano 2012	Cross-sectional (French schools)	6,590 primary-school children	Rhinoconjunctivitis	OR 1.17 per 10 µg/m ³ classroom FA
Krzyżanowski 1990	Cohort (Arizona households)	298 households	Decreased PEFR in children	Significant dose-response ≥0.06 ppm
Wang 2015	Cross-sectional occupational	74 exposed + 63 controls	Sperm quality	Lower concentration/motility

Study	Design	Population	Endpoint	Key effect estimate
	(China)			ty in exposed
Morales 2025	Ambient-air monitoring (Puerto Rico)	Urban residential sites	Cancer prioritisation	Formaldehyde top-5 air priority
Chu 2025	Cross-sectional occupational	Formaldehyde-exposed Taiwanese workers	Respiratory symptoms, spirometry	Elevated symptoms at 0.2–0.5 ppm

Appendix D. Mode of Action and Key Events

The EPA IRIS 2024 Toxicological Review articulated a mode-of-action (MOA) framework for formaldehyde carcinogenicity based on a series of key events derived from inhalation studies in rats, monkeys, and humans. The central mechanistic claim is that inhaled formaldehyde produces upper respiratory tumours via a predominantly mutagenic mode of action driven by DNA-protein crosslink formation, impaired repair, and secondary proliferative response. The key events are summarised below.

Key event	Process	Evidence source
1	Covalent reaction of formaldehyde with respiratory epithelial biomolecules	DNA, protein, glutathione conjugates formed in nasal epithelium at doses from endogenous baseline upward
2	DNA-protein crosslink formation	Quantified in rat and monkey nasal epithelium by $^{13}\text{CD}_2$ isotope-dilution MS (Swenberg 2013)
3	Replication-fork collapse, transcription blockade, and replication-dependent double-strand breaks	Mulderrig 2024, Fujihara 2024
4	Mutagenesis via translesion synthesis and mis-repair of double-strand breaks	Signatures in rat nasal tumour genomes; Polθ-dependent translesion synthesis (Niu 2025)
5	Sustained cytotoxicity and regenerative cell proliferation in the anterior nasal epithelium	Documented by BrdU labelling in 13-week inhalation studies at 6 and 15 ppm
6	Clonal expansion of initiated cells and progression to squamous cell carcinoma	Two-year rat inhalation bioassays at 6 and 15 ppm
7	(Leukaemia pathway) Haematopoietic stem cell injury from elevated endogenous/exogenous formaldehyde pools	Niu 2025 (ADH5/ALDH2/POLQ); Hauptmann 2009 (NCI embalmers)

The MOA framework supports two implications adopted in the 2024 EPA IRIS assessment. First, because the first four key events involve DNA damage, the low-dose cancer extrapolation retains its linear default without a threshold. Second, because key events 2–4 are maturationally sensitive (developing tissue has higher cell-proliferation rate and more active transcription and replication), age-dependent adjustment factors (ADAFs) are applied: tenfold for ages 0–2, threefold for ages 2–16, and no adjustment for ages 16+.

Appendix E. Authoritative Agency Assessment Summaries

This appendix compiles structured summaries of the five principal authoritative agency assessments relied on throughout this review. Each summary captures the assessment scope, the health-based values derived, the identification of critical effects, and the principal contributions to the contemporary understanding of formaldehyde toxicity. Where a quantitative value is presented the primary source is the agency assessment as cited in the bibliography.

E.1 EPA IRIS 2024 Toxicological Review of Formaldehyde (Inhalation)

The 2024 IRIS assessment is the first comprehensive update of the EPA IRIS formaldehyde file since the noncancer and cancer files were separately developed in 1990–1991. The assessment was released as the final Toxicological Review in August 2024 following internal review drafts in 2010 and 2022 and an interagency science discussion draft in 2024. The document is 927 pages long and identifies the following health-based values: (i) a noncancer chronic inhalation reference concentration (RfC) of 0.007 mg/m³, derived as the median of three organ-specific RfCs for rhinoconjunctivitis, pulmonary function, and asthma control with high overall confidence; (ii) an inhalation unit risk (IUR) for cancer of 1.1×10^{-5} per µg/m³, derived from the Beane Freeman (2013) NCI cohort by linear low-dose extrapolation with age-dependent adjustment factors (ADAFs) for the mutagenic mode of action at the nasal site; (iii) retention without change of the 1990 chronic oral RfD of 0.2 mg/kg-day based on reduced weight gain and histopathology in rats with a NOAEL of 15 mg/kg-day and a composite uncertainty factor of 100; and (iv) a weight-of-evidence conclusion that formaldehyde is “carcinogenic to humans by the inhalation route of exposure” supported by three independent causal judgments for nasopharyngeal cancer, sinonasal cancer, and myeloid leukemia.

Key methodological features of the 2024 IRIS assessment include the use of benchmark dose (BMD) modelling at the point of departure for each candidate critical endpoint, explicit probabilistic consideration of inter-individual variability (UFH = 3 rather than the default 10 because of strong human data), and application of age-dependent adjustment factors for early-life exposures in the cancer IUR derivation. The 2024 document departed from the 2010 external review draft by setting aside the Conolly CIIT biologically-based dose-response (BBDR) model in favour of direct epidemiology-based cancer risk estimation, citing concerns about identifiability of key BBDR parameters and about the extent to which the rat-based model could be reliably extrapolated to the human nasopharyngeal tumour context.

E.2 NTP 15th Report on Carcinogens (2021)

The NTP Report on Carcinogens is a Congressionally mandated biennial science-based list of substances identified as either “known to be a human carcinogen” or “reasonably anticipated to be a human carcinogen.” The 15th Edition (RoC 15) was published in 2021 and retained the listing of formaldehyde as “known to be a human carcinogen” first adopted in the 12th RoC (2011). Formaldehyde was originally listed in the 2nd Annual Report on Carcinogens (1981) as “reasonably anticipated”; the 2011 upgrade

followed the IARC 2006 Group 1 designation and the publication of the NCI cohort analyses by Hauptmann (2009) and Beane Freeman (2009).

The NTP RoC profile describes the causal relationship with three cancer sites: nasopharyngeal cancer (supported by the Vaughan 2000 case-control study with OR 13.3, 95% CI 2.5–70 for definite exposure, and by the Beane Freeman NCI cohort with an exposure-response gradient by peak exposure), sinonasal cancer (supported by the Luce 2002 pooled analysis of twelve case-control studies and by the Roush 1987 mortality study of embalmers), and myeloid leukaemia (supported by the Beane Freeman 2009 NCI industrial cohort, the Hauptmann 2009 NCI embalmers case-control study, and the Zhang 2009 meta-analysis). The NTP profile also includes detailed exposure characterisation for the general population and occupational workforce, a comprehensive survey of mechanistic studies through 2020, and an assessment of cancer burden in the US population at current exposures.

E.3 ATSDR Toxicological Profile tp111 (1999)

The ATSDR Toxicological Profile for Formaldehyde (tp111.pdf, 468 pages, 1999) is the comprehensive US Public Health Service review of all health effects from formaldehyde. Although published twenty-five years before the 2024 IRIS assessment, the ATSDR profile remains the principal source for acute and intermediate-duration health-based values and for non-cancer endpoints at the full-dose response. The document derives acute (1–14 day), intermediate (15–364 day), and chronic (≥ 365 day) inhalation Minimal Risk Levels (MRLs) at 0.04, 0.03, and 0.008 ppm respectively; oral MRLs at the same three durations at 0.3, 0.3, and 0.2 mg/kg-day. The nine main chapters cover Public Health Statement, Health Effects by Route and Duration, Chemical and Physical Information, Production/Import/Use/Disposal, Potential for Human Exposure, Analytical Methods, Regulations and Advisories, References, and Glossary.

E.4 WHO Environmental Health Criteria 89 (1989)

WHO Environmental Health Criteria (EHC) monograph 89 was published in 1989 under the UNEP/ILO/WHO International Programme on Chemical Safety (IPCS) and remains the principal WHO-level comprehensive assessment of formaldehyde. The 219-page document is organised into twelve chapters covering identity and physical/chemical properties; analytical methods; sources of environmental pollution; environmental transport, distribution, and transformation; environmental concentrations and human exposure; metabolism and kinetics in laboratory animals and humans; effects on organisms in the environment; effects on experimental animals and in vitro test systems; effects on humans; evaluation of human health risks; conclusions; and recommendations. Although superseded for quantitative risk assessment by the 2010 WHO Indoor Air Quality guideline of 0.1 mg/m³ (30-minute average) and by the EPA IRIS 2024 RfC, the EHC monograph remains the definitive global consensus reference for formaldehyde environmental fate, exposure, and non-cancer health effects through the end of the 1980s.

E.5 NIOSH Pocket Guide to Chemical Hazards, entry npgd0293

The NIOSH Pocket Guide entry for formaldehyde (npgd0293, last reviewed 30 October 2019) consolidates the occupational-hygiene essentials. Key fields include the NIOSH Recommended Exposure Limit (REL) of Ca TWA 0.016 ppm with a ceiling of 0.1 ppm for 15-minute exposures, with the chemical classified as an Appendix A potential occupational carcinogen. The OSHA Permissible Exposure Limit is referenced as 29 CFR 1910.1048 TWA 0.75 ppm and STEL 2 ppm. The Immediately Dangerous to Life or Health (IDLH) value is 20 ppm. Physicochemical data given include molecular weight 30.03 g/mol, boiling point -6°F , freezing point -134°F , vapour pressure greater than 1 atmosphere (reflecting that pure formaldehyde is a gas at 20°C), ionisation potential 10.88 eV, lower and upper explosive limits 7% and 73% by volume, and miscibility with water. Incompatibilities include strong oxidisers, alkalis and acids, phenols, and urea, with specific mention that formaldehyde reacts with hydrochloric acid to form bis-chloromethyl ether (BCME) — a potent lung carcinogen. Exposure routes are inhalation, ingestion, skin contact, and eye contact. Target organs are the eyes and the respiratory system; the entry flags nasal cancer as the site of concern.

Appendix F. Glossary and Acronyms

Acronym / term	Definition
ADAF	Age-dependent adjustment factor. A multiplier applied to the unit risk estimate to account for increased cancer risk from early-life exposure when the mode of action is mutagenic; EPA defaults are 10 for ages 0–2, 3 for ages 2–16, and 1 for 16+.
ADH5	Alcohol dehydrogenase 5 (S-hydroxymethyl-glutathione dehydrogenase, formerly ADH III). The principal enzyme oxidising formaldehyde-glutathione adducts to formate.
AICIS	Australian Industrial Chemicals Introduction Scheme (formerly NICNAS).
ALDH2	Mitochondrial aldehyde dehydrogenase. Oxidises acetaldehyde, formaldehyde, and other aldehydes directly to the corresponding carboxylic acid.
AMeD syndrome	Aplastic anaemia, Mental retardation, Dwarfism syndrome — caused by compound loss-of-function in ALDH2 and ADH5 (or FANC genes), producing elevated endogenous formaldehyde burden.
ATSDR	Agency for Toxic Substances and Disease Registry (US Public Health Service).
BMD / BMDL	Benchmark dose / benchmark dose lower bound — the 95% lower confidence bound on the dose producing a specified benchmark response.
CAS	Chemical Abstracts Service registry number.
CFD	Computational fluid dynamics — used to model inhaled-vapour deposition in the respiratory tract.
DPC	DNA-protein crosslink — the principal genotoxic lesion produced by formaldehyde.
DSSTox	Distributed Structure-Searchable Toxicity database (EPA CompTox Chemicals Dashboard).
DTXSID	DSSTox Substance Identifier.
EHC	WHO IPCS Environmental Health Criteria monograph series.
FA / FOSN / UF / MUF / PF	Formaldehyde; Formaldehyde-emitting structural nailings; urea-formaldehyde resin; melamine-urea-formaldehyde; phenol-formaldehyde.
IARC	International Agency for Research on Cancer (WHO).
IDLH	Immediately Dangerous to Life or Health (NIOSH).
IRIS	Integrated Risk Information System (EPA).
IUR	Inhalation Unit Risk — cancer risk per $\mu\text{g}/\text{m}^3$ continuous lifetime exposure.
LC ₅₀ / LD ₅₀	Median lethal concentration / dose.

Acronym / term	Definition
LOAEL / NOAEL	Lowest observed adverse effect level / no observed adverse effect level.
MOA	Mode of action.
MRL	Minimal Risk Level (ATSDR).
NCI	National Cancer Institute.
NIOSH	National Institute for Occupational Safety and Health.
NPC	Nasopharyngeal carcinoma.
NTP	National Toxicology Program (US).
OSHA	Occupational Safety and Health Administration (US).
PEL	Permissible Exposure Limit (OSHA).
PMC / PMCID / PMID	PubMed Central / PMC ID / PubMed ID.
REL	Recommended Exposure Limit (NIOSH).
RfC / RfD	Reference concentration (inhalation) / reference dose (oral) — EPA non-cancer benchmarks.
RoC	NTP Report on Carcinogens.
SMR	Standardised mortality ratio.
STEL	Short-term exposure limit — typically a 15-minute ceiling.
TLV	Threshold Limit Value (ACGIH).
TWA	Time-weighted average.
UFC	Composite uncertainty factor.
WoE	Weight of evidence.

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